GLOBAL OLYMPICS SPORTS PERFORMANCE ANALYSIS

Project Documentation

Data Analytics with Tableau Public

Prepared by: Abhivant Gupta
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An interactive exploration of Olympic medal performance by country, year, sport, medal type and gender.

1. Problem Statement

The project explores Olympic medal performance using connected datasets in Tableau Public. It presents multiple views of medal results so that users can compare countries, examine changes across Games years, view geographic medal distribution, and investigate medal categories, sports and gender.

Objectives

- Connect the Olympic CSV files in a single Tableau data source using relationships.
- Create visualizations for medals by country, year, geographic location, medal type, sport and gender.
- Combine the worksheets into an Olympic dashboard.
- Provide a Tableau Story entry point for presenting the analysis in a guided sequence.
- Document the data model, visual design and current project outputs.

2. Dataset and Data Source

The Tableau data source is named “medal_results+”. The Data Source screen shows four CSV tables connected around medal_results.csv: alltime.csv, athletes.csv and games_summary.csv. A requirements.txt file is also visible in the Files pane, but it is a project/deployment file rather than a data table.

File	Role / visible contents	Evidence from the workbook
medal_results.csv	Central medal-results table in the relationship model.	Shown as the central table on the Data Source canvas.
alltime.csv	All-time country medal information used for country-level analysis.	Country (Alltime.Csv) is used in the country bar chart.
athletes.csv	Athlete attributes, including Athlete Id, Name, Gender, Age At Peak, Country and Noc Code.	26 fields and 8,000 rows are shown in the data preview.
games_summary.csv	Games-level summary information, including year and country-related fields.	Year (Games Summary.Csv) is used in the year trend.
requirements.txt	Python dependency list for the application environment.	Visible in the Tableau Files pane; not part of the relationship model.

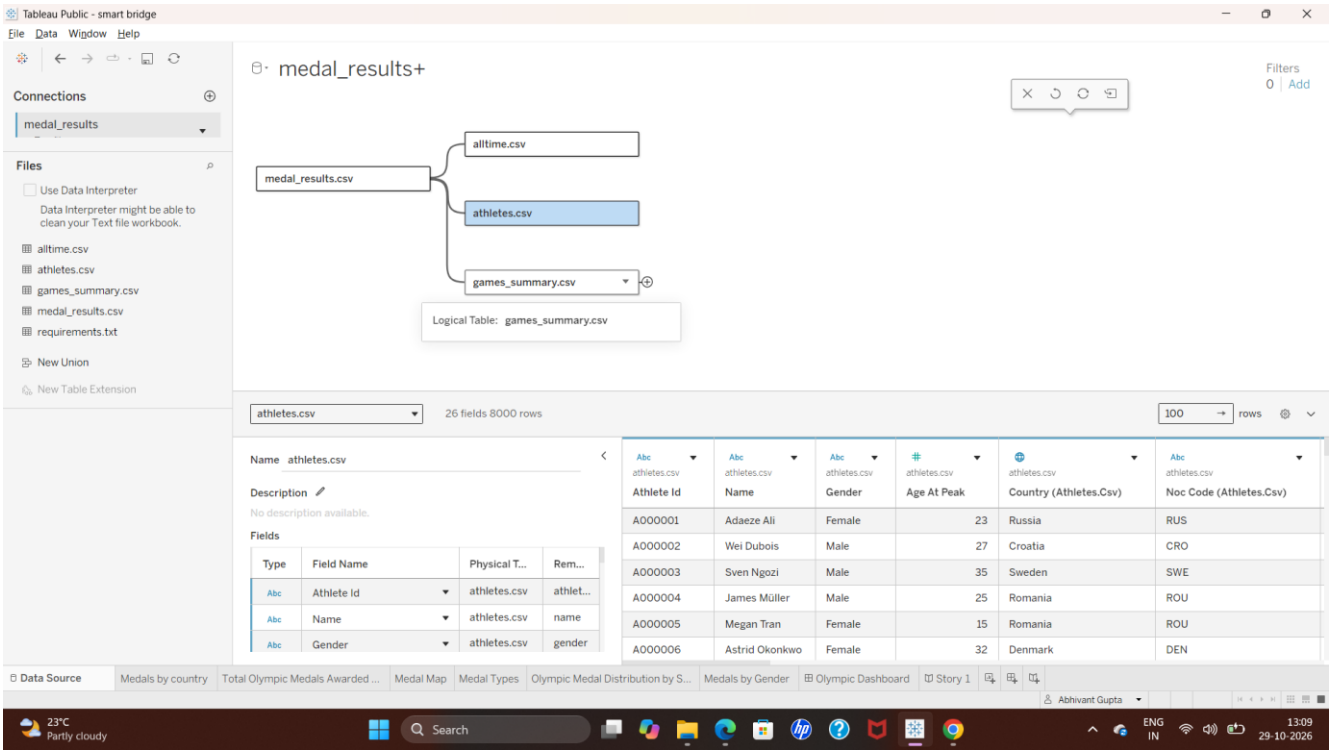


Figure 1. Tableau data source and connected CSV tables.

3. Data Preparation and Model

The screenshots show the four CSV tables arranged in Tableau's logical data model. The medal_results.csv table is placed centrally, with alltime.csv, athletes.csv and games_summary.csv connected as related logical tables. This relationship-based model keeps the source tables separate while making their fields available for analysis.

Preparation steps represented in the workbook

1. Loaded the CSV files into Tableau Public through the Files connection pane.
2. Placed medal_results.csv on the data-source canvas as the central logical table.
3. Added alltime.csv, athletes.csv and games_summary.csv as related tables.
4. Inspected the athletes.csv preview, where Tableau displays 26 fields and 8,000 rows.
5. Used fields from the related tables in the worksheets, including country, year, gender, sport and medal measures.

Note: The screenshots do not expose the exact relationship key fields, null-handling rules, data-cleaning operations or extract/live-connection setting. Those implementation details are therefore not asserted in this document.

4. Data Visualizations

The workbook contains six analysis worksheets. The following descriptions are based on the visible worksheet tabs, chart titles, axes, legends and marks in the supplied screenshots.

Worksheet	Chart form	What it communicates
Medals by country	Horizontal bar chart	Compares total medals for the displayed top 10 countries. A country filter is visible on the right.
Total Olympic Medals Awarded by year	Line chart	Shows the total medals awarded across the Games years represented in games_summary.csv.
Medal Map	Filled map	Uses geographic country shading to compare total medal counts; the legend shown ranges from 11 to 735.
Medal Types	Three side-by-side bar views	Compares Bronze, Gold and Silver medal totals by country.
Olympic Medal Distribution by Sport	Treemap	Uses differently sized rectangles to show medal distribution across sports.
Medals by Gender	Two-category bar chart	Compares the displayed medal totals for Female and Male athletes.

4.1 Medals by Country

The horizontal bar chart ranks ten displayed countries by SUM(Total Medals). The visible country labels include Australia, China, France, Germany, Great Britain, Italy, Japan, Russia, South Korea and the United States. A country filter is available for interactive selection.

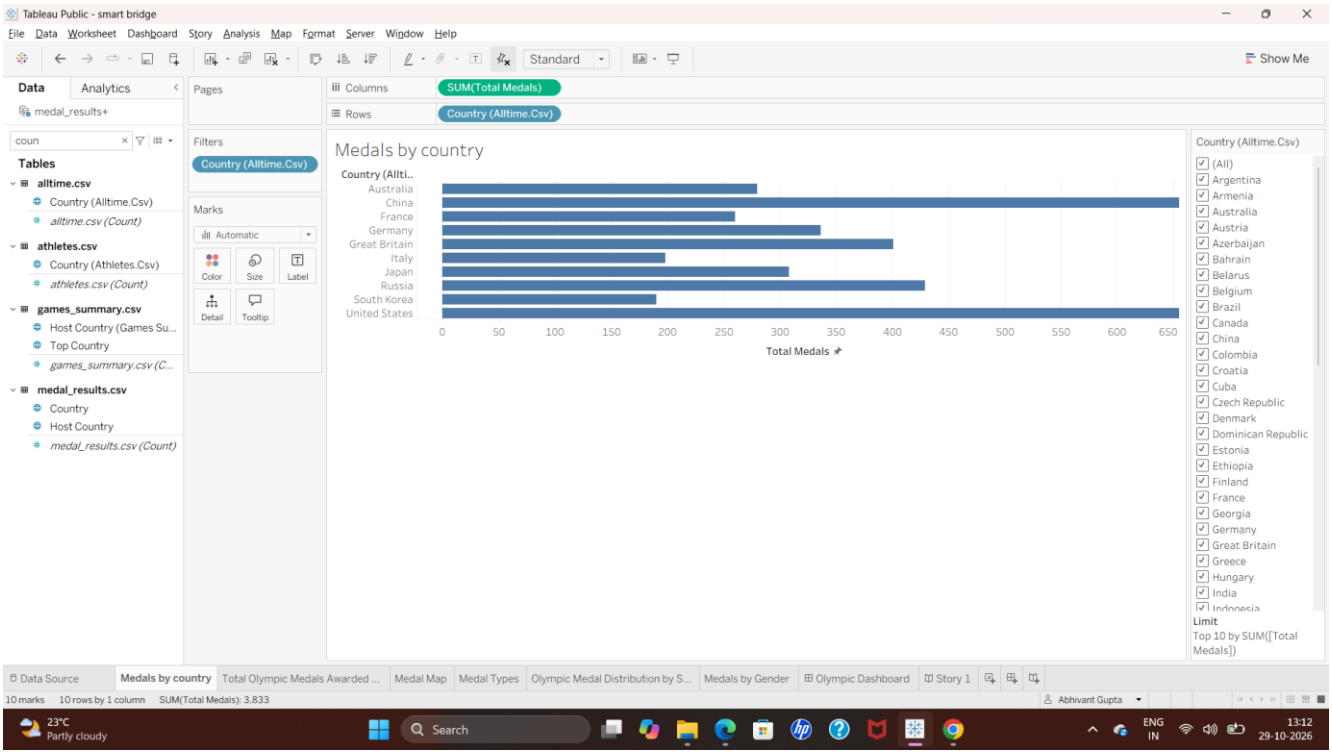


Figure 2. Medals by country (top 10 view).

4.2 Total Olympic Medals Awarded by Year

The line chart plots Year (Games Summary.Csv) against SUM(Total Medals Awarded). The visible marks cover 18 year values, with the displayed line varying across the years.

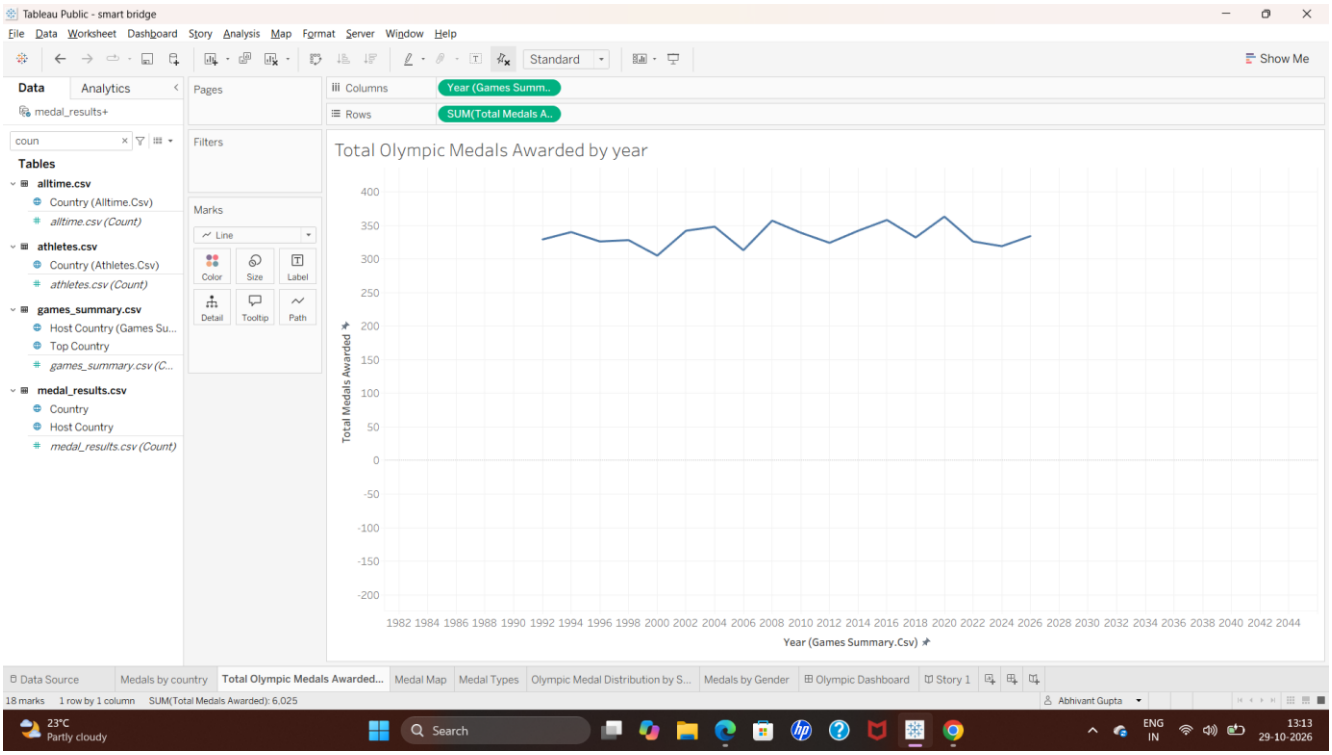


Figure 3. Total Olympic medals awarded by year.

4.3 Medal Map

The filled map uses generated longitude and latitude with Country as the geographic detail. Colour intensity represents SUM(Total Medals). The tooltip shown for Canada reports 158 total medals.

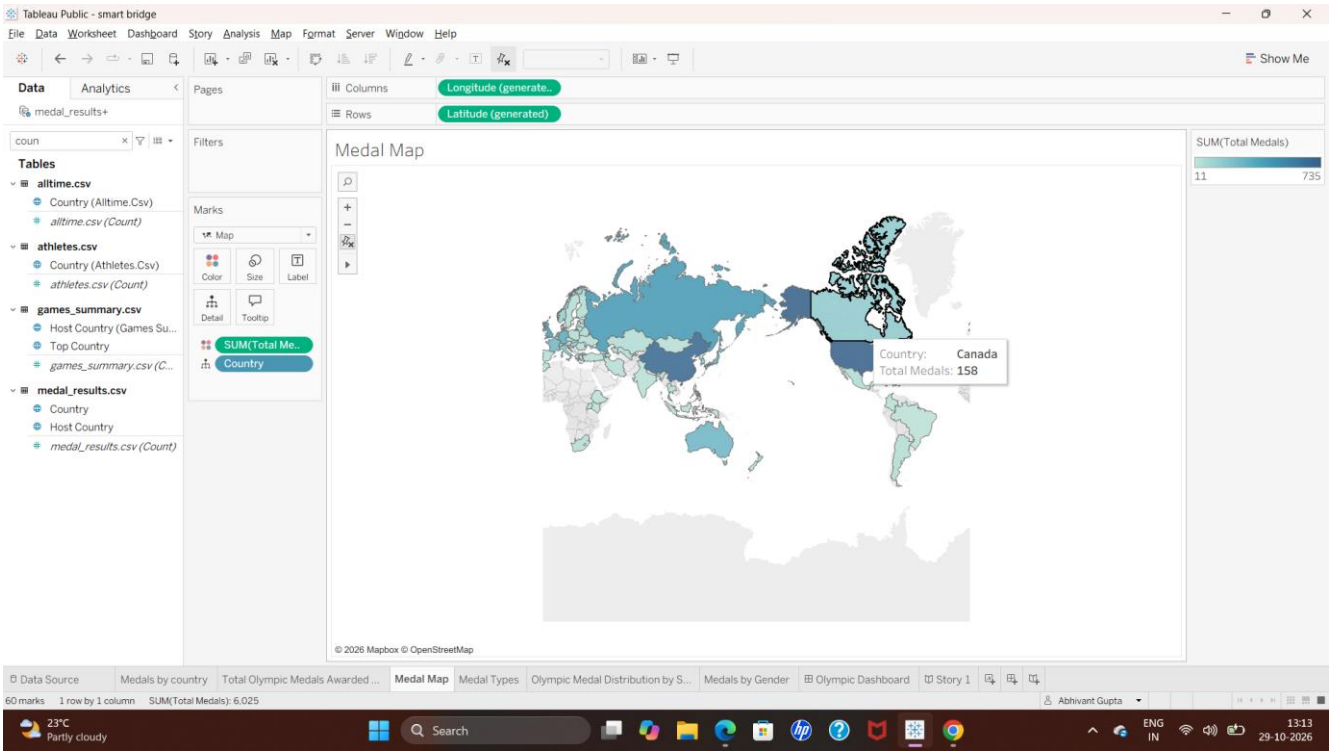


Figure 4. Geographic distribution of total medals.

4.4 Medal Types

This worksheet presents separate horizontal bar views for Bronze, Gold and Silver by country. It allows the medal categories to be compared side by side.

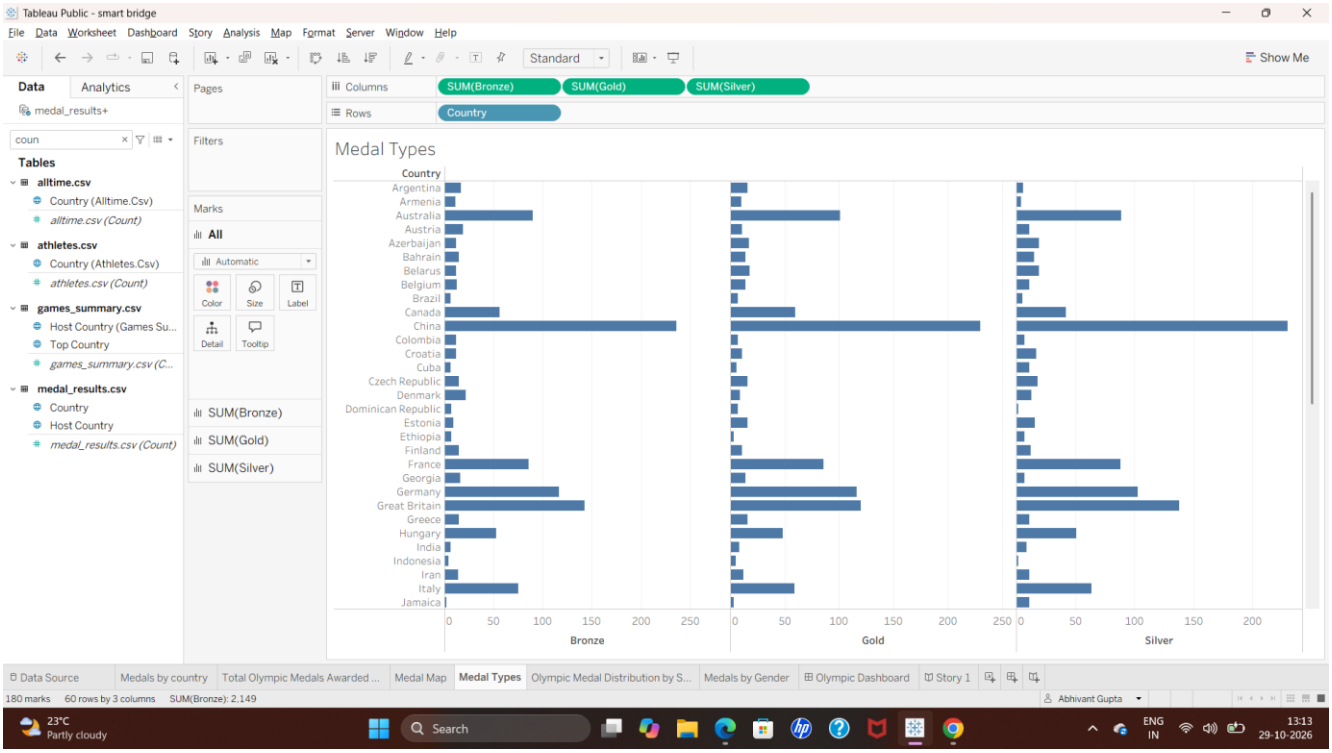


Figure 5. Bronze, gold and silver medals by country.

4.5 Olympic Medal Distribution by Sport

The treemap displays sports as rectangles. The rectangle area and colour encode the medal measure, making it possible to compare the relative contribution of sports at a glance. Labels visible in the screenshot include Handball, Shooting, Table Tennis, Skateboarding, Rugby Sevens, Boxing, Surfing, Cycling, Football/Soccer, Athletics, Weightlifting, Sailing, Hockey (Field), Basketball, Swimming, Diving, Archery, Judo, Wrestling, Fencing, Rowing, Sport Climbing and other sports.

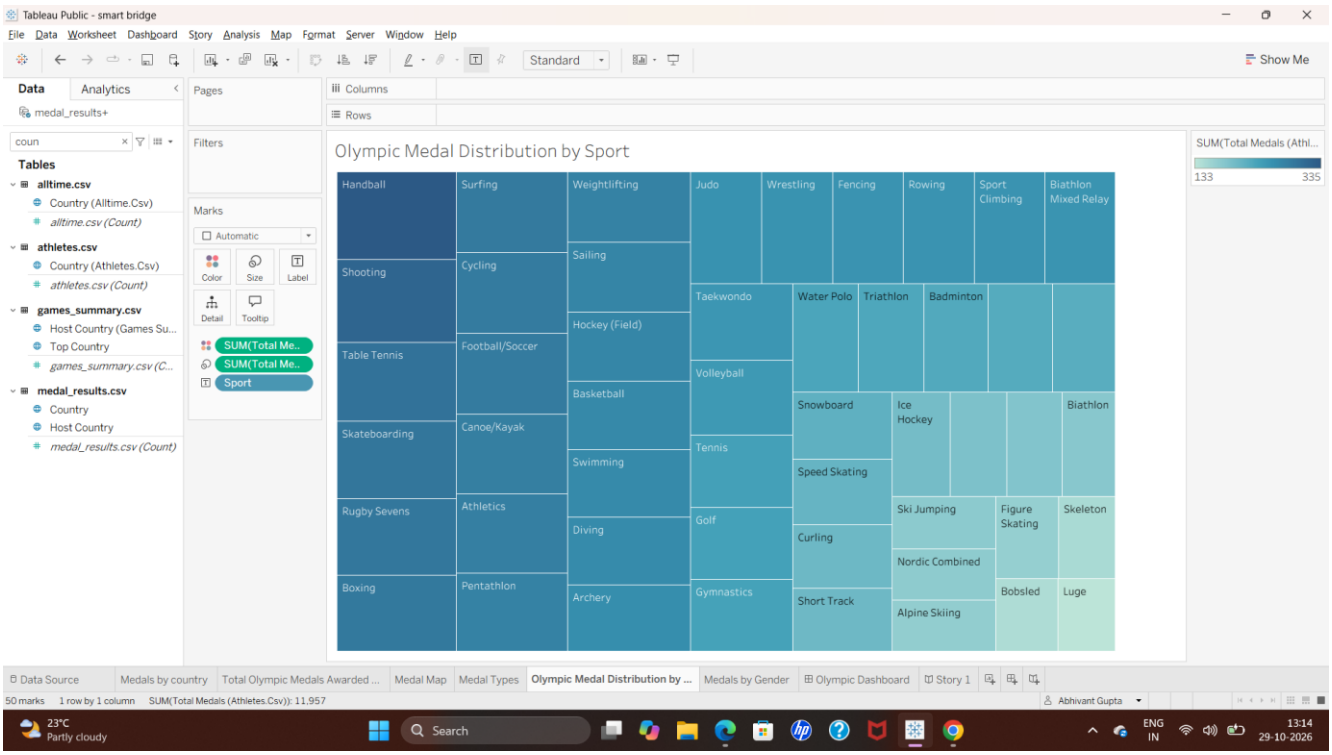


Figure 6. Treemap of Olympic medal distribution by sport.

4.6 Medals by Gender

The chart compares Female and Male categories using the Gender field. The displayed measure is SUM(Total Medals (Athletes.Csv)); the screenshot status bar reports 11,957 for the worksheet total.

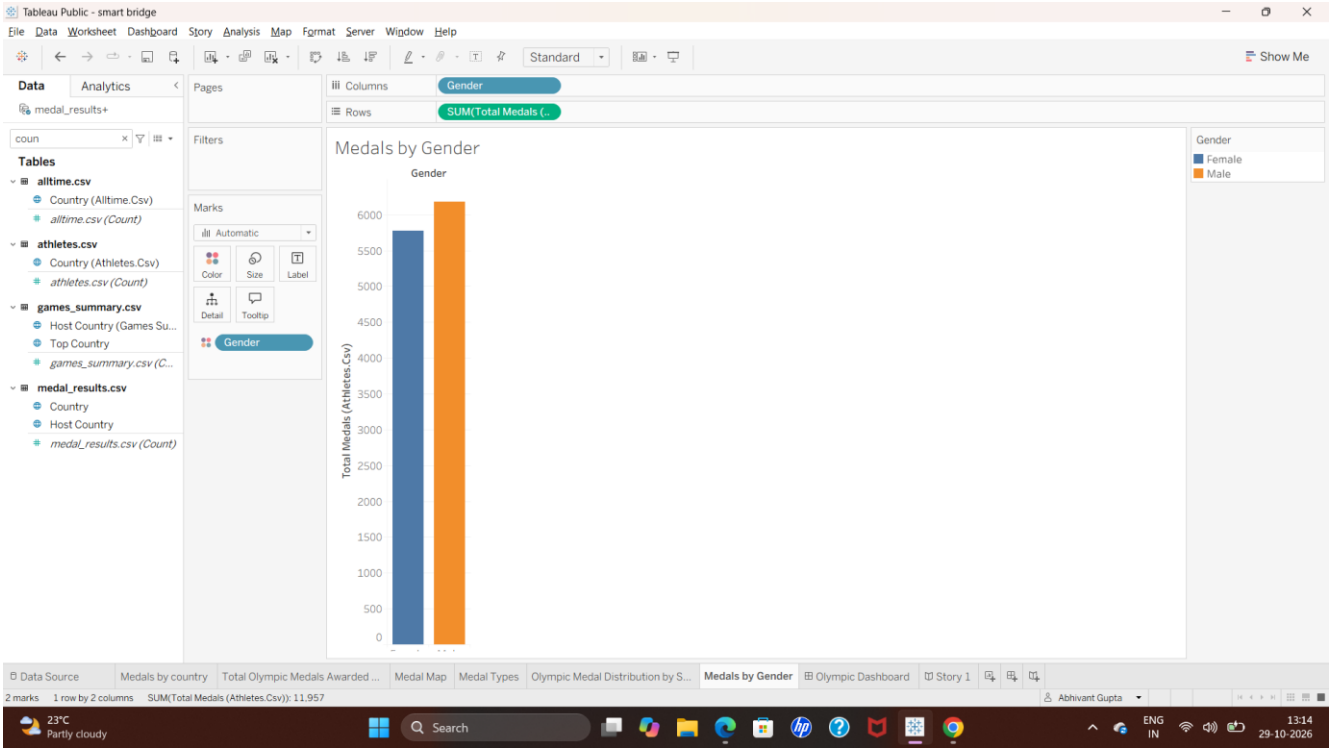


Figure 7. Medal totals by gender.

5. Olympic Dashboard

The Olympic Dashboard combines the project worksheets into one Tableau view. The supplied screenshot shows a dashboard title and a tiled layout with the following visual areas:

- Medals by Gender in the upper-left area.
- Medals by Country in the upper-middle area.
- Medal Map in the upper-right area.
- Medal Types in the lower-left area.
- Total Olympic Medals Awarded by Year in the lower-middle area.
- Olympic Medal Distribution by Sport in the lower-right area.

The dashboard also displays a Country filter and legends for the medal and gender measures. The Dashboard pane shows the size setting as Automatic. The screenshot demonstrates a compact overview in which multiple views can be read together; specific filter-action behaviour is not confirmed by the supplied image.

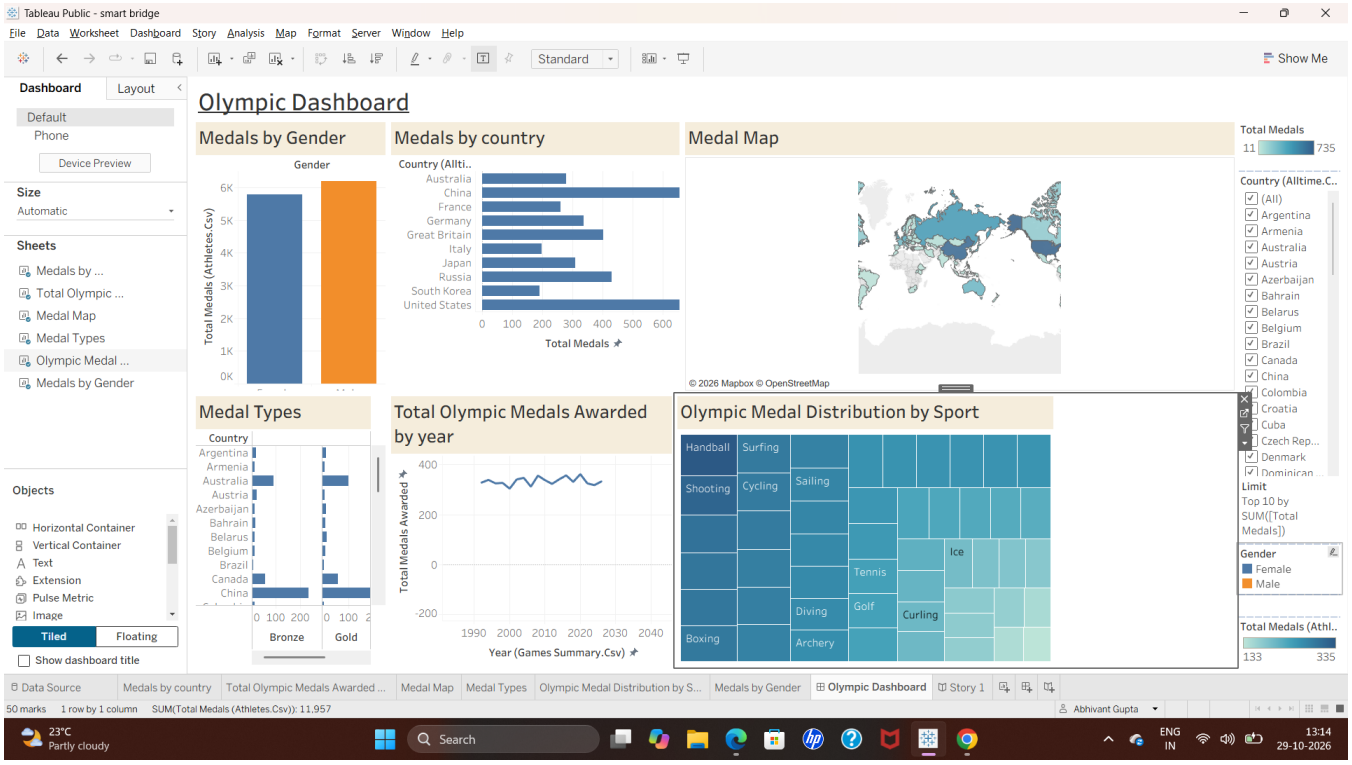


Figure 8. Olympic Dashboard in Tableau Public.

6. Tableau Story

A Story tab named “Story 1” is visible in the workbook tab bar. This confirms that a Tableau Story has been created in the workbook. The supplied screenshots do not show the story-point captions or the contents of individual story points, so a scene-by-scene narrative is not documented here.

Suggested documentation to complete this section

- Capture the Story 1 view with all story points and captions visible.
- Record the order and purpose of each story point.
- Add one or two sentences explaining the main insight communicated by each scene.

7. Workbook Checks and Performance

The screenshots confirm that the worksheets and dashboard are present in Tableau Public and that the charts render with visible marks and legends. They do not include timed load tests, a performance recorder output, or a documented optimization comparison.

Check	What can be confirmed from the screenshots
Data source	Four CSV logical tables are visible in the data model.
Athlete data preview	26 fields and 8,000 rows are displayed for athletes.csv.
Worksheets	Six analysis worksheets are visible in the workbook tabs.
Dashboard	An Olympic Dashboard worksheet is present and populated with multiple views.
Story	A Story 1 tab is visible; story-point details are not shown.
Load-time performance	Not measured in the supplied evidence.

8. Project Links and Publishing

The Tableau Public share links previously supplied for this project are listed below. They can be used to open the published story and dashboard.

Tableau Public Story: https://public.tableau.com/views/smartbridge_17931188449870/Story1

Tableau Public Dashboard: https://public.tableau.com/views/smartbridge_17931188449870/OlympicDashboard

The screenshots show the workbook in Tableau Public. A GitHub repository URL, deployed web-app URL and demo-video URL were not included in the materials used for this document.

9. Conclusion and Learnings

The project brings together Olympic medal data in Tableau Public and presents it through country, year, map, medal-type, sport and gender views. The dashboard places these analyses in one workspace to support comparison across several dimensions of Olympic performance.

Key project outcomes

- Connected four CSV data tables in a Tableau logical data model.
- Created six worksheets covering complementary views of medal performance.
- Built an Olympic Dashboard containing the six analysis views.
- Created a Tableau Story tab as an additional presentation format.

Skills demonstrated

- Connecting and relating multiple CSV files in Tableau Public.
- Building horizontal bar charts, line charts, filled maps, treemaps and medal-category comparisons.
- Using country, year, gender, sport and medal measures in analytical views.
- Arranging multiple worksheets into a dashboard.

10. Further Improvements

The following additions would make the documentation and project handover more complete:

- Add the exact relationship keys and a short explanation of the data model.
- Document any calculated fields, filters, parameters and dashboard actions used.
- Add screenshots of the Story 1 points and their captions.

- Record repeatable load-time tests for each worksheet and the dashboard.
- Add the GitHub repository and deployed Flask application links if web integration is part of the final submission.